

3. Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$

(6)



7. Prove by mathematical induction that, for $n \in \mathbb{N}$

$$\begin{pmatrix} -5 & 9 \\ -4 & 7 \end{pmatrix}^n = \begin{pmatrix} 1 - 6n & 9n \\ -4n & 1 + 6n \end{pmatrix}$$

(6)



8. (i) Prove by induction that for $n \in \mathbb{Z}^+$

$$\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix} \quad (6)$$

(ii) Prove by induction that for $n \in \mathbb{Z}^+$

$$f(n) = 4^{n+1} + 5^{2n-1}$$

is divisible by 21

(6)

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- 6.** (a) Prove by induction that for all positive integers n ,

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1) \quad (6)$$

- (b) Use the standard results for $\sum_{r=1}^n r^3$ and $\sum_{r=1}^n r$ to show that for all positive integers n ,

$$\sum_{r=1}^n r(r+6)(r-6) = \frac{1}{4}n(n+1)(n-8)(n+9) \quad (4)$$

- (c) Hence find the value of n that satisfies

$$\sum_{r=1}^n r(r+6)(r-6) = 17 \sum_{r=1}^n r^2 \quad (5)$$